



Mechanical Behaviour of Engineering Materials

Groover: Chapters 2 and 5



The student:

- knows the nature of materials;
- knows the basic structure of metals;
- knows the basic structure of polymers.





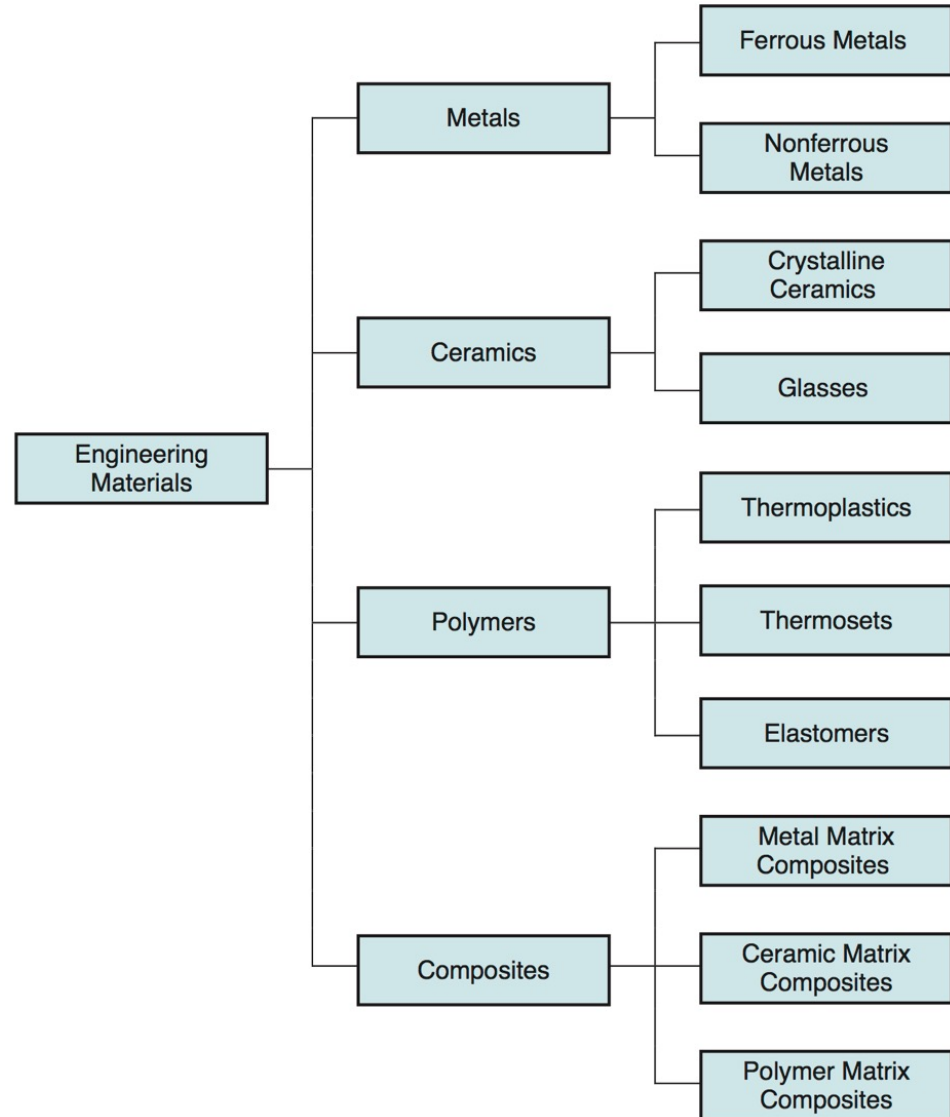
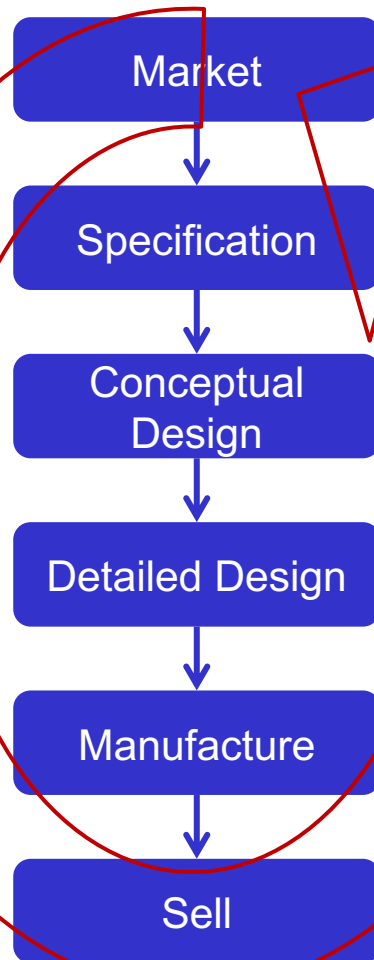
IMPORTANCE OF MATERIALS IN MANUFACTURING

Manufacturing is a transformation process:

- It is the material that is transformed,
- And it is the behavior of the material when subjected to the forces, temperatures, and other parameters of the process that determines the success of the operation.



PRODUCT DESIGN AND MANUFACTURING





The elements can be grouped into families and relationships established between and within the families by means of the Periodic Table.

1 IA																	18 VIIIA
1 H	2 He																
3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
11 Na	12 Mg	3 IIIV	4 IVB	5 VB	6 VIB	7 VIIB	8	9 VII	10	11 IB	12 IIB	13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
55 Cs	56 Ba	57-71	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
87 Fr	88 Ra	89-103	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Uut	114 Fl	115 Uup	116 Lv	117 Uus	118 Uuo
6 La	57 Ce	58 Pr	59 Nd	60 Pm	61 Sm	62 Eu	63 Gd	64 Tb	65 Dy	66 Ho	67 Er	68 Tm	69 Yb	70 Lu			
7 Ac	89 Th	90 Pa	91 U	92 Np	93 Pu	94 Am	95 Cm	96 Bk	97 Cf	98 Es	99 Fm	100 Md	101 No	102 Lr			

Alkali Metals
 Alkali Earth Metals
 Transition Metals
 Other Metals
 Metalloids

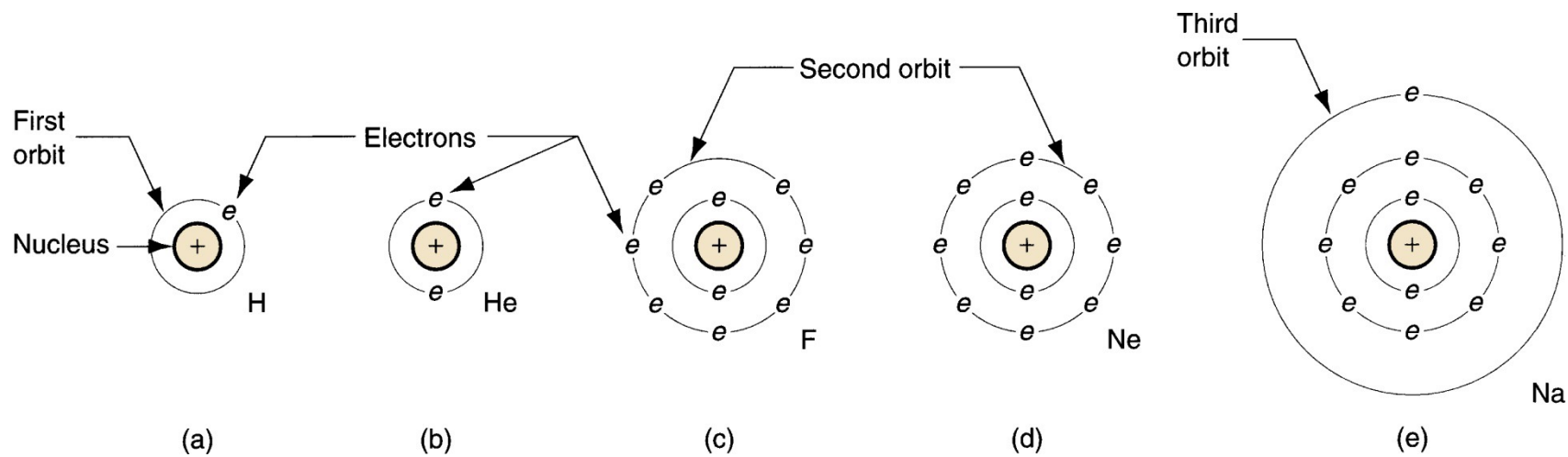
Other Non Metals
 Halogens
 Noble Gases
 Lanthanides & Actinides



ATOMIC STRUCTURE AND THE ELEMENTS

The basic structural unit of matter is the atom:

- Each atom is composed of a positively charged nucleus, surrounded by a sufficient number of negatively charged electrons so the charges are balanced;
- More than 100 elements, and they are the chemical building blocks of all matter.



(a) Hydrogen, (b) helium, (c) fluorine, (d) neon, and (e) sodium



BONDING BETWEEN ATOMS AND MOLECULES

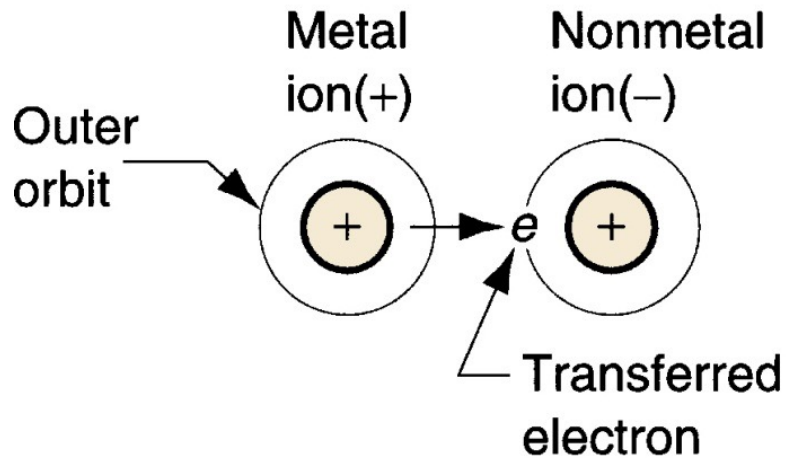
Atoms are held together in molecules by various types of bonds:

1. **Primary bonds** - Characterized by strong atom-to-atom attractions that involve exchange of valence electrons. Types: ***Ionic***, ***Covalent***, ***Metallic***.
2. **Secondary bonds** - Generally associated with attraction between molecules, no transfer or sharing of electrons. Types: *Dipole forces*, *London forces*, *Hydrogen bonding*.

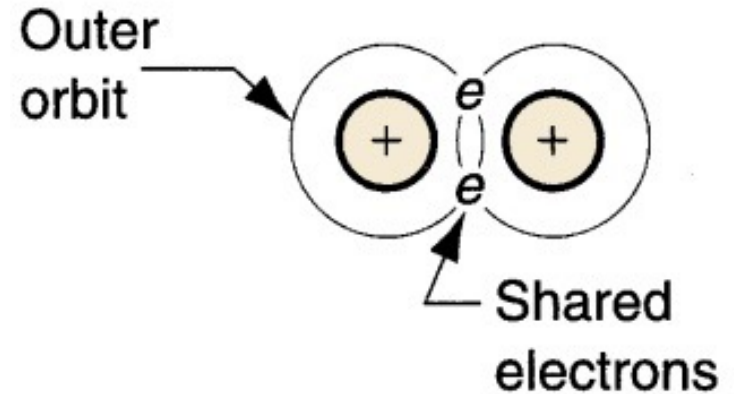
Primary bonds are much stronger than secondary bonds.



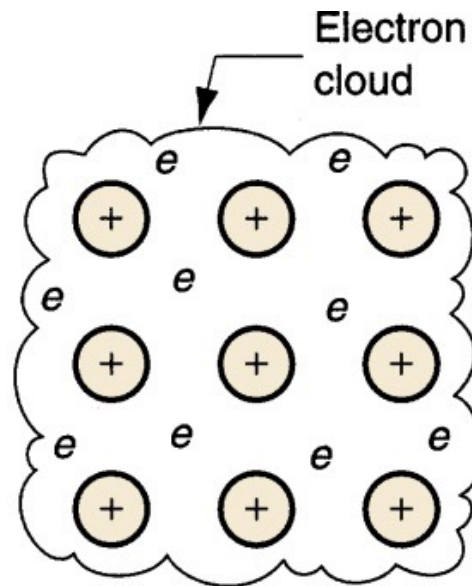
Ionic Bond



Covalent Bond



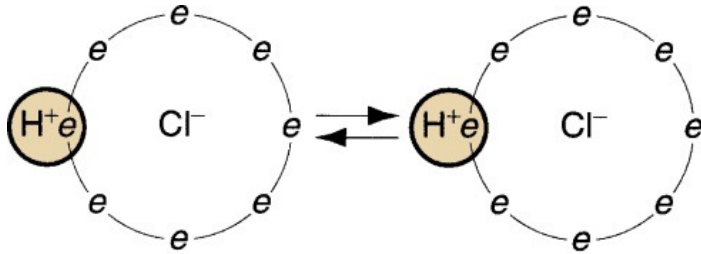
Metallic Bond



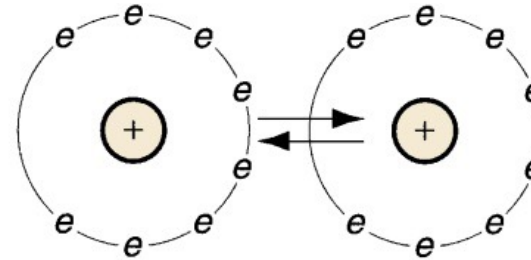


SECONDARY BONDS

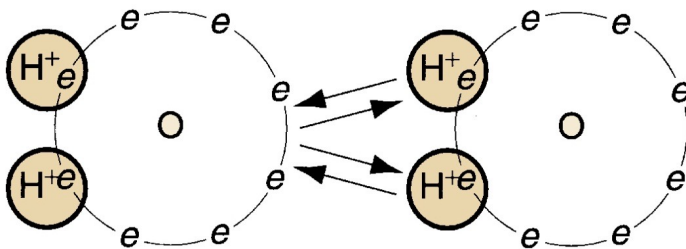
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Dipole Forces: Arise in a molecule comprised of two atoms with equal and opposite electrical charges. Each molecule therefore forms a dipole that attracts other molecules.



London Forces: Attractive force between non-polar molecules, i.e., atoms in molecule do not form dipoles. However, temporary dipoles form when more electrons are on one side.

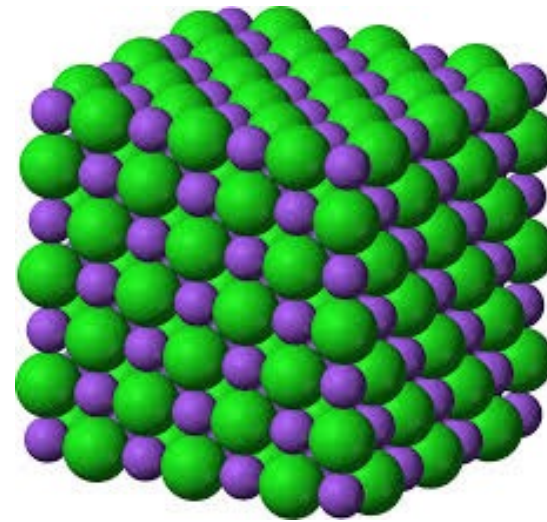
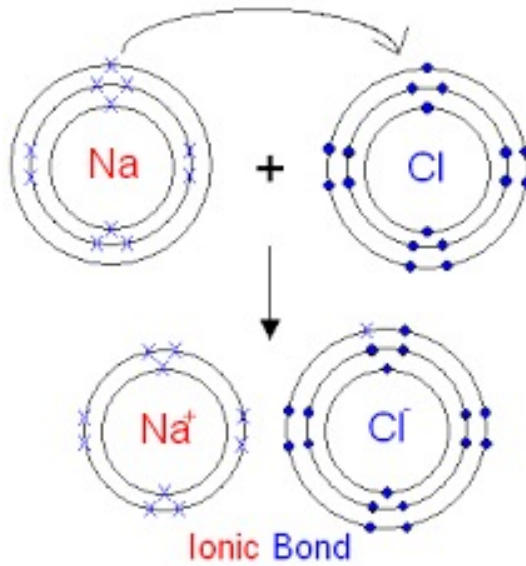


Hydrogen Bonding: Occurs in molecules containing hydrogen atoms covalently bonded to another atom. Since electrons to complete shell of hydrogen atom are aligned on one side of nucleus, opposite side has a net positive charge that attracts electrons in other molecules.



EXAMPLE – CHLORIDE SODIUM

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MACROSCOPIC STRUCTURES OF MATTER

Atoms and molecules are the building blocks of a more macroscopic structure of matter.

When materials solidify from the molten state, they tend to close ranks and pack tightly, arranging themselves into one of two structures:

- Crystalline;
- Non-crystalline.



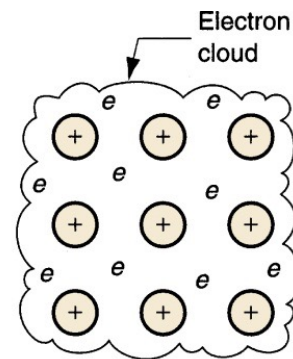
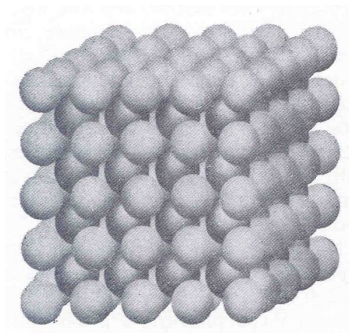
- **Metals:** crystalline structure in the solid state. High hardness, electrical and thermal conductivity, opaqueness, reflectivity.
- **Ceramics:** either crystalline or non-crystalline structure (amorphous). High hardness, stiffness, brittleness, electrical insulation properties, refractoriness, chemical inertness.
- **Polymers:** either a glassy structure or mixture of glassy and crystalline structure. Low density, high electrical resistivity, low thermal conductivity. Strength and stiffness vary widely.



All metals, except for mercury, are in solid phase at room temperature.

At solid (and ideal) phase, atoms are located in a regular structure (**crystalline structure**):

- Nodes $\Rightarrow$ Atoms;
- Internal electrons close to nuclei;
- External electrons can move.

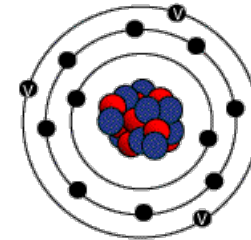




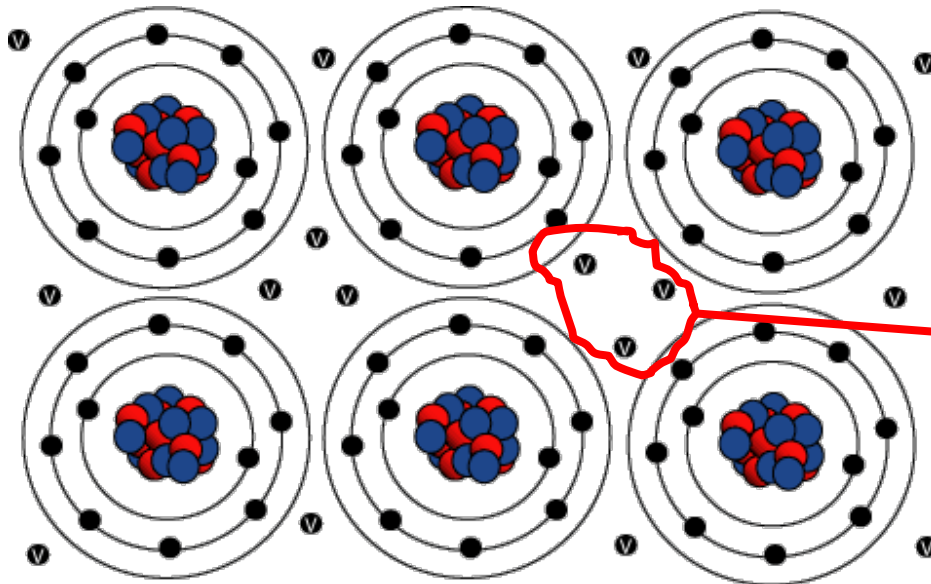
EXAMPLE – ALUMINUM

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Aluminum atomic structure (13)



3 valence electrons

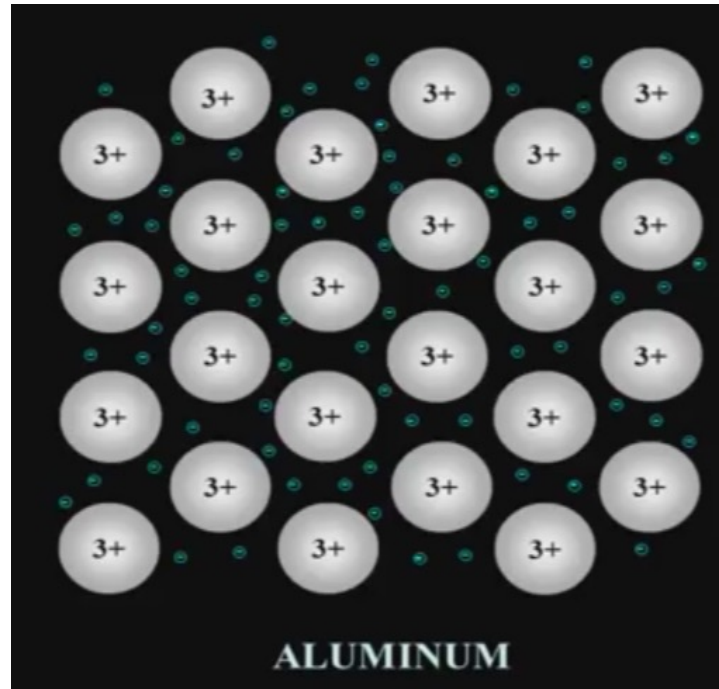


Valence electrons
shared in the cloud.



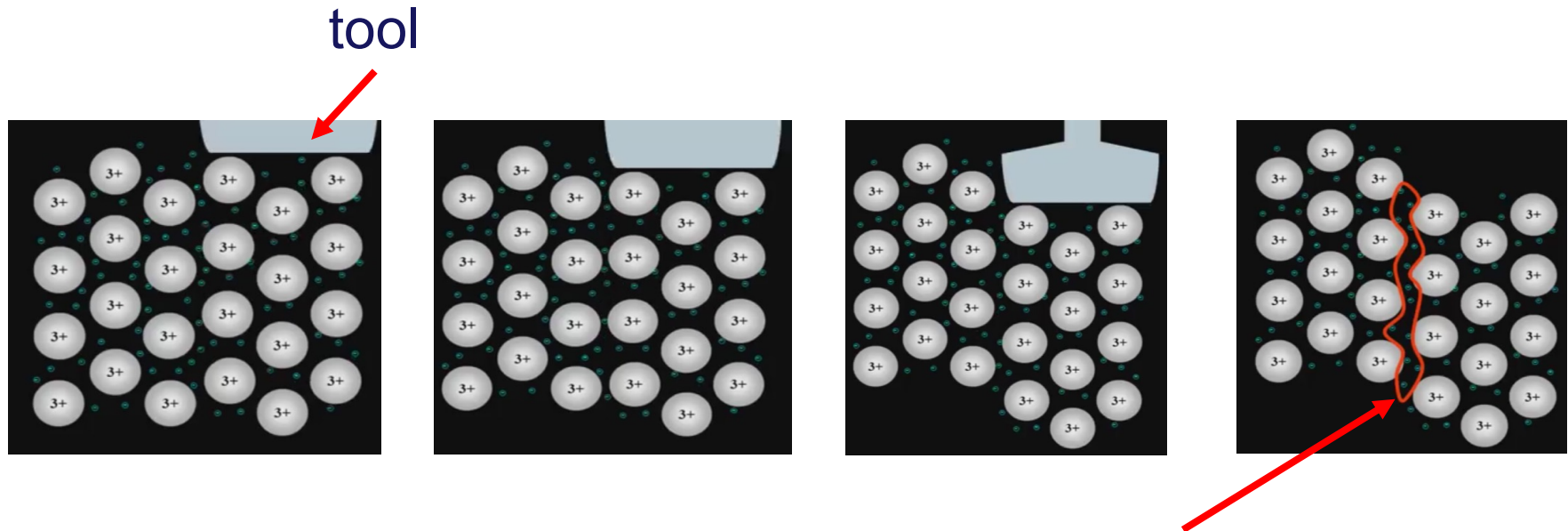
EXAMPLE – METALLIC BOND

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PROPERTIES FROM METALLIC BOND: MALLEABILITY

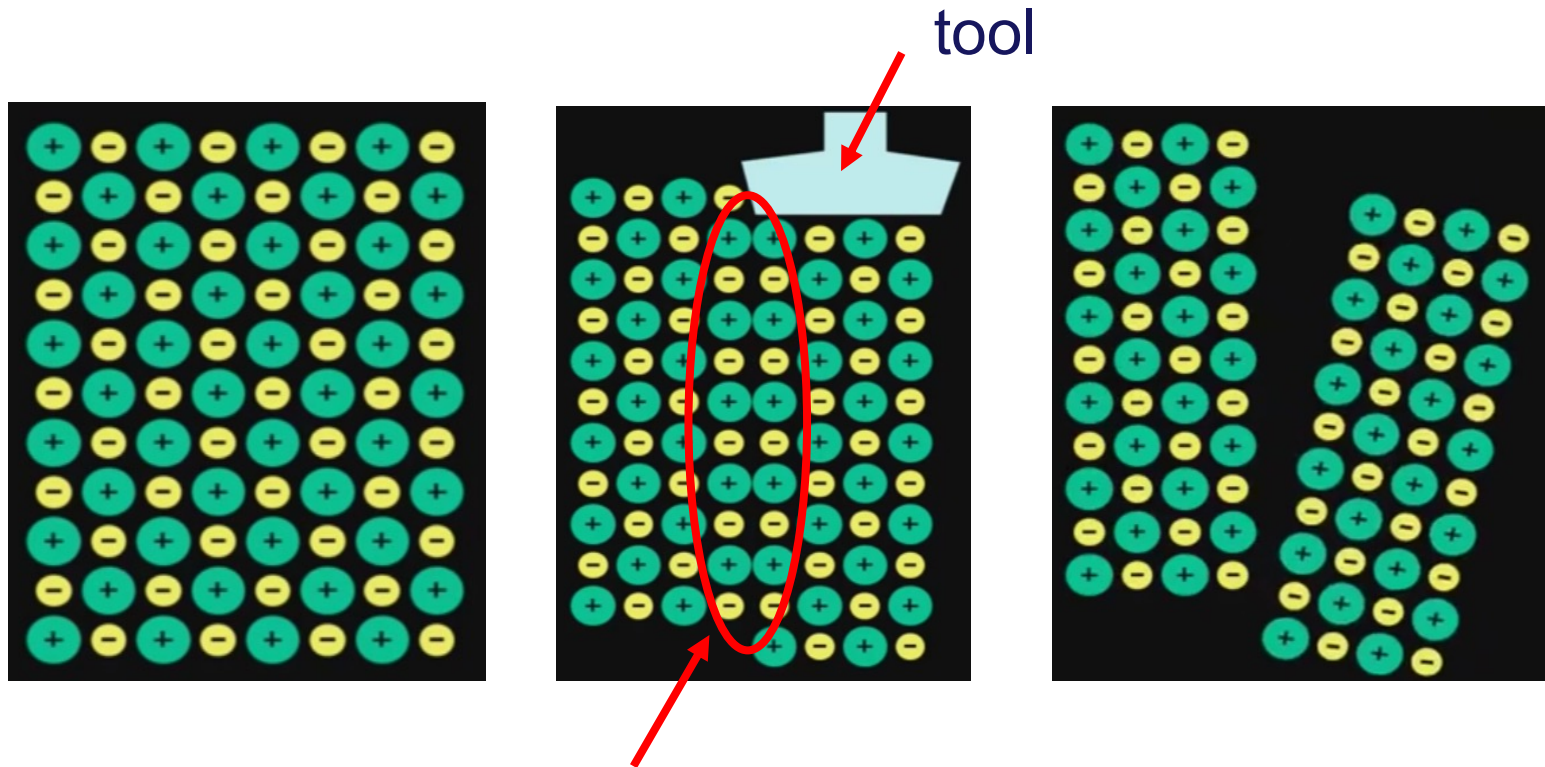


The delocalized electrons
are a constant presence.



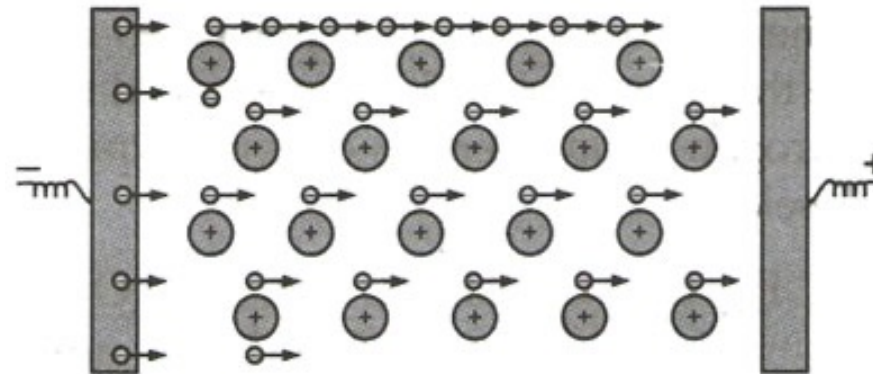
PROPERTIES FROM METALLIC BOND: MALLEABILITY

Ionic structures are not malleable.



PROPERTIES FROM METALLIC BOND

- *Electrical conductivity*: delocalized electrons also carry electronic current and heat because of their ability to move through crystals.



Electrical conductivity in metal crystals

PROPERTIES FROM METALLIC BOND

- *Lustre*: electrons readily absorb and re-emit visible frequency photons, giving metals their characteristic lustre. Surface roughness affects this property. What if you scratch?



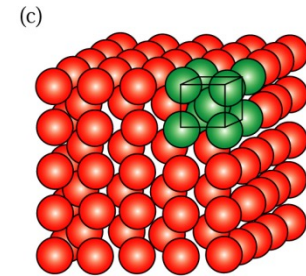
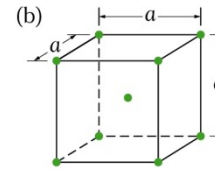
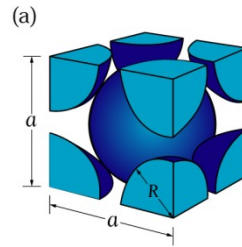
Cloud gate (A. Kapoor, 2006)
Millennium Park - Chicago
Stainless Steel



BASIC CRYSTAL STRUCTURES (UNIT CELLS)

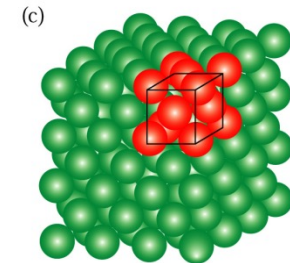
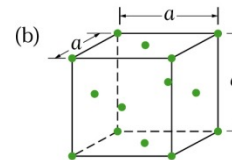
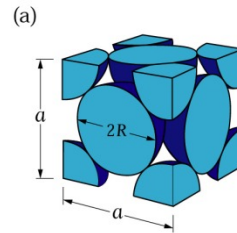
Body-Centered Cubic (BCC)

Metals: Iron (Fe- α , Fe- δ), Chromium (Cr), Molybdenum (Mo), Tungsten (W), ...



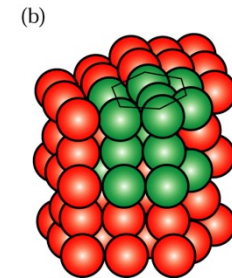
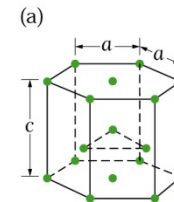
Face-Centered Cubic (FCC)

Metals: Iron (Fe- γ), Aluminium (Al), Copper (Cu), Nickel (Ni), Silver (Ag), Gold (Au), ...



Hexagonal Close-Packed (HCP)

Metals: Magnesium (Mg), Titanium (Ti), Zinc (Zn), ...



Iron: Fe- α , Fe- δ , Fe- γ . The appearance of more than one type of crystal structure is known as **allotropism** or **polymorphism** (meaning “many shapes”).



MECHANICAL PROPERTIES OF MATERIALS

Mechanical properties determine the behavior of a material when subjected to mechanical stresses.

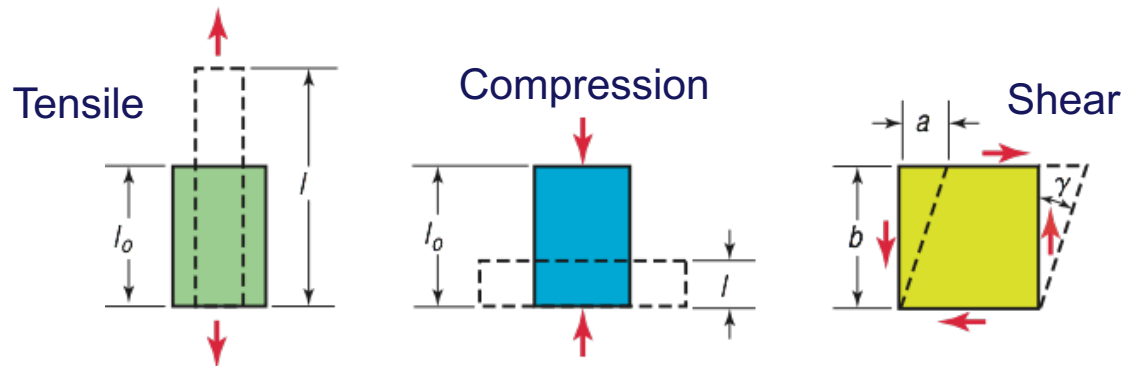
Dilemma: mechanical properties desirable to the designer, such as high strength, usually make manufacturing more difficult.

- The manufacturing engineer should consider the design viewpoint.
- The designer should be aware of the manufacturing viewpoint.

MECHANICAL PROPERTIES OF MATERIALS

Three types of static stresses to which materials can be subjected:

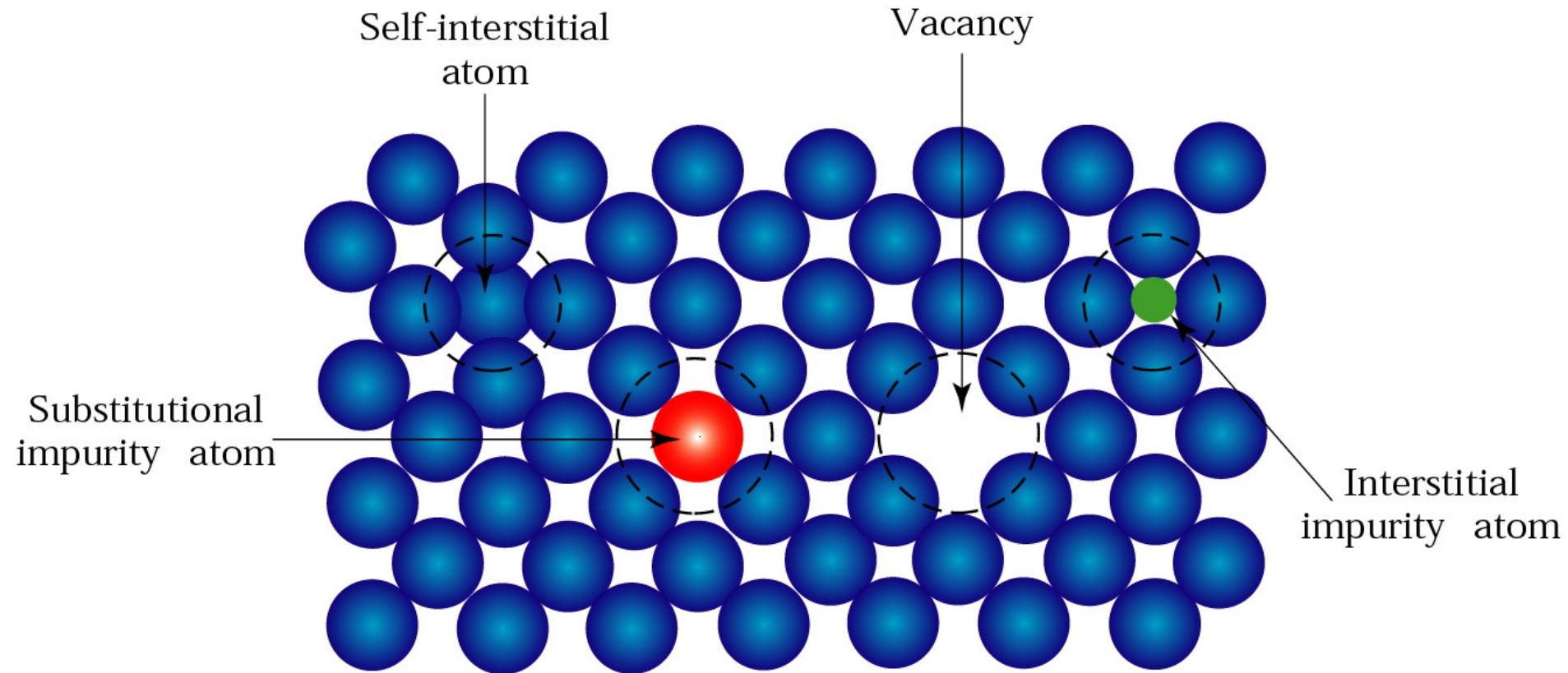
- *Tensile* - tend to stretch the material;
- *Compressive* - tend to squeeze it;
- *Shear* - tend to cause adjacent portions of material to slide against each other.

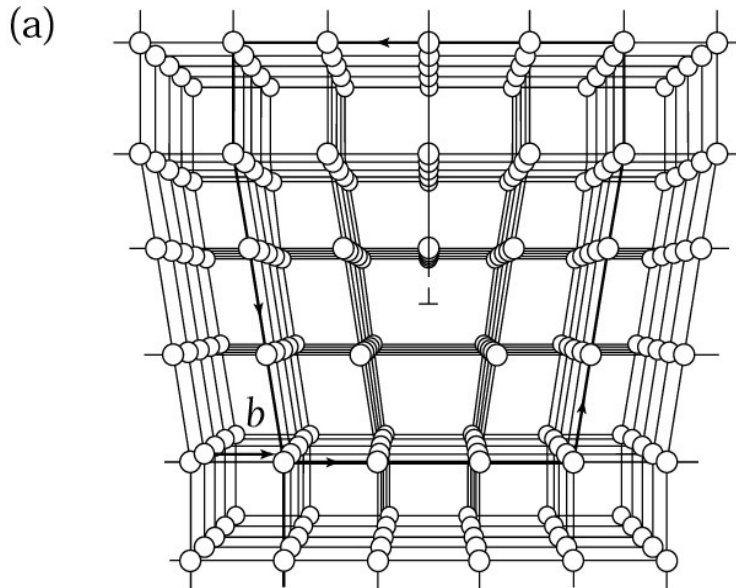


Stress - Strain curve: basic relationship that describes mechanical properties for all three types.



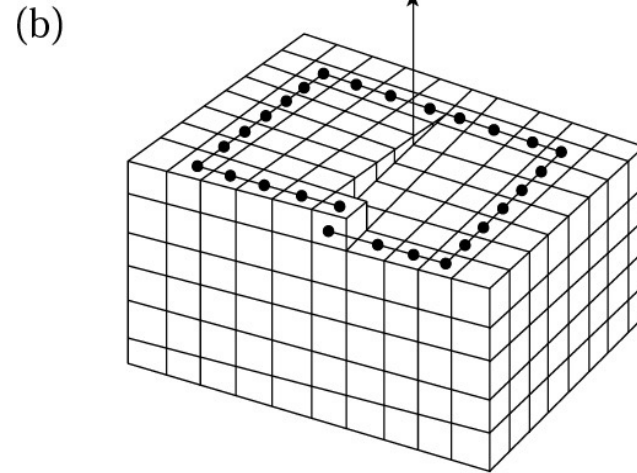
- In theory metals should resist more than they do in reality.
- The presence of imperfections reduces the resistance of metals:
 - **Point defects:** interstitial atom, vacancy, impurity atom;
 - **Linear defects:** dislocations;
 - **Surface defects:** grain boundaries, phase boundaries;
 - **Volume defects:** voids, inclusions, other phases, cracks.





Edge dislocation

A linear defect at the edge of an extra plane of atoms.

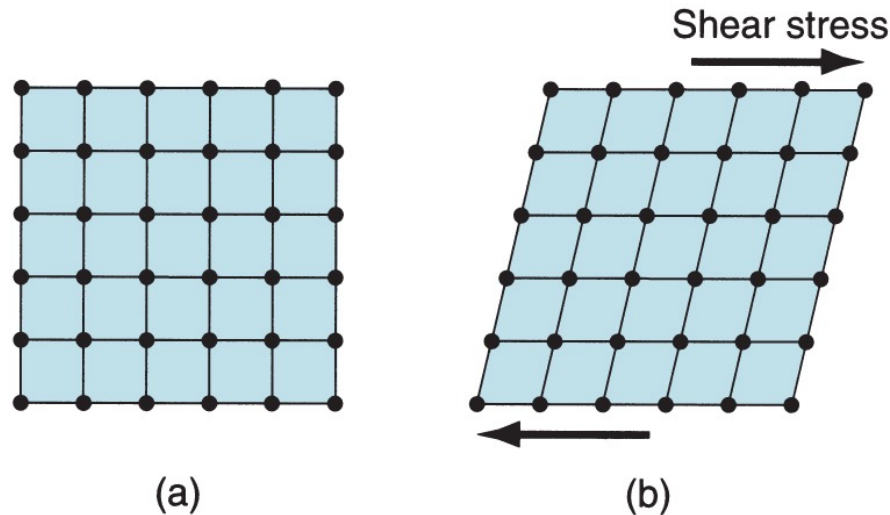


Screw dislocation

A helical defect in a three-dimensional lattice of atoms. Screw dislocations are so named because the atomic planes form a spiral ramp.



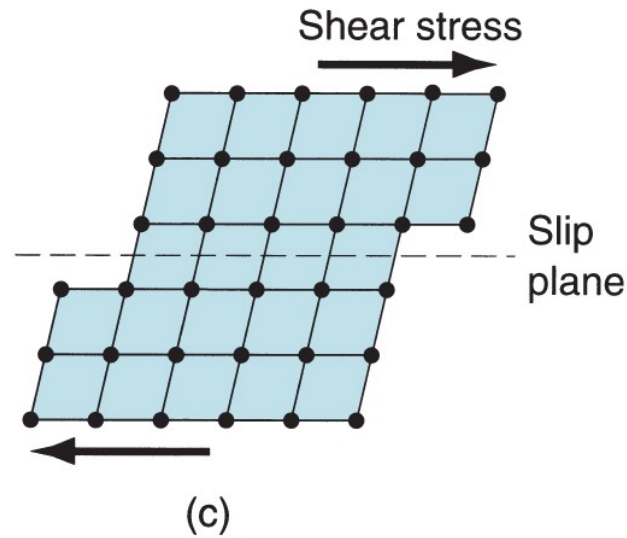
When a crystal experiences a gradually increasing stress, it first deforms elastically.



Deformation of a crystal structure: (a) original lattice: (b) elastic deformation, no permanent change in positions of atoms.



If the stress is higher than forces holding atoms in their lattice positions, then a permanent shape change occurs.

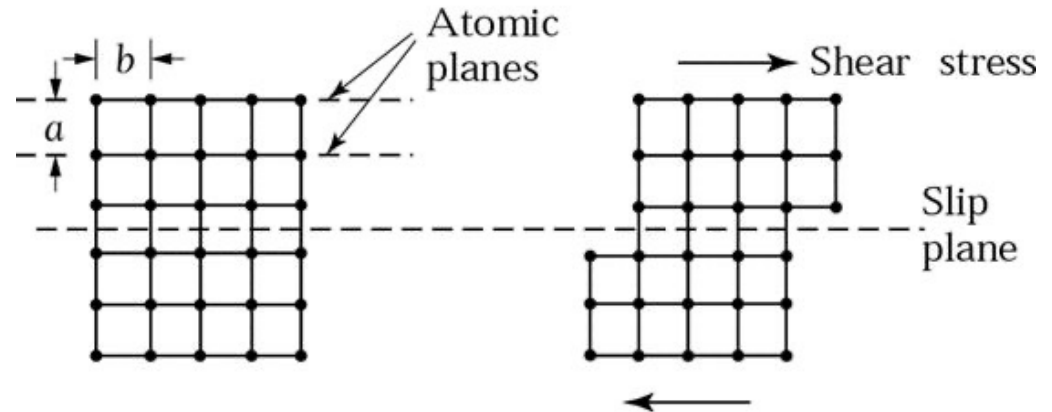


Plastic deformation (slip), in which atoms in the crystal lattice structure are forced to move to new “homes”.

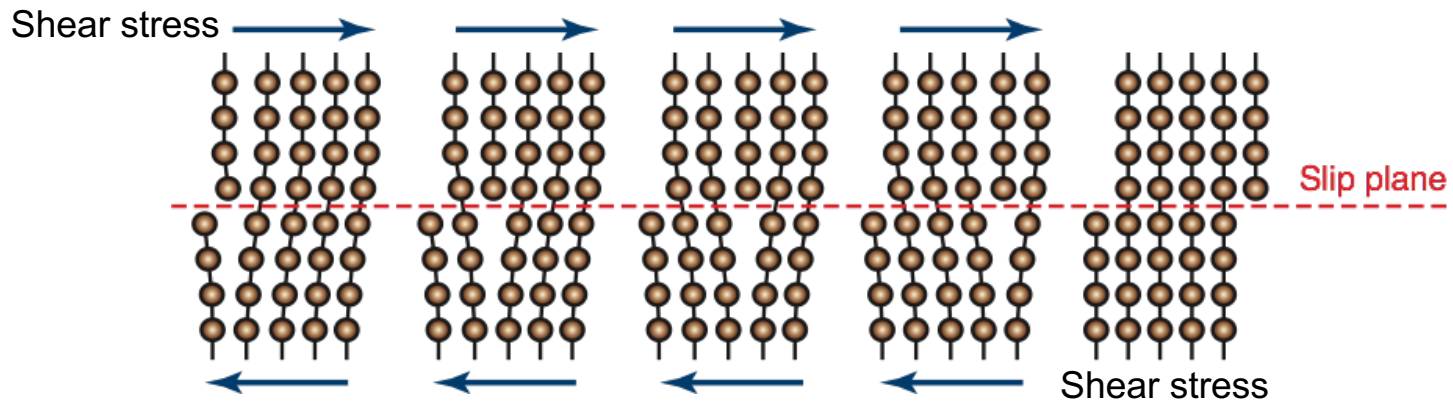


DEFORMATION IN METALLIC CRYSTALS

Deformation of a crystal structure



Effect of dislocations in the lattice structure under stress.





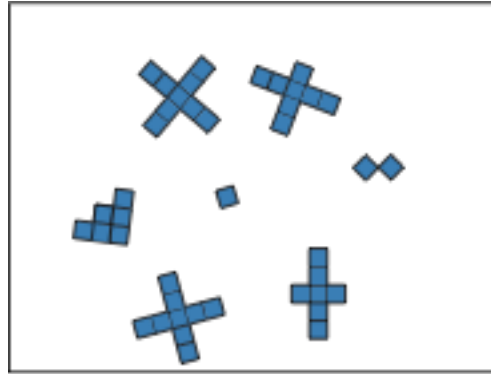
POLYCRYSTALLINE NATURE OF METALS

A block of metal may contain millions of individual crystals, called grains. Such a structure is called ***polycrystalline***:

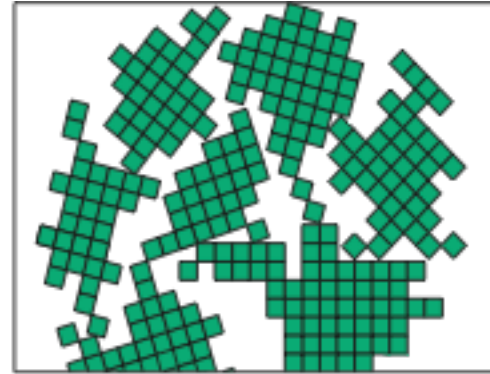
- Each grain has its own unique lattice orientation;
- But collectively, the grains are randomly oriented in the block.

How do polycrystalline structures form?

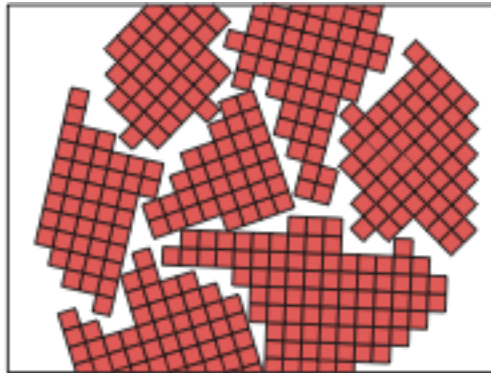
- As a volume of metal cools from the molten state and begins to solidify, individual crystals nucleate at random positions and orientations throughout the liquid;
- These crystals grow and finally interfere with each other, forming at their interface a surface defect - a grain boundary, which is a transition zone, perhaps only a few atoms thick.



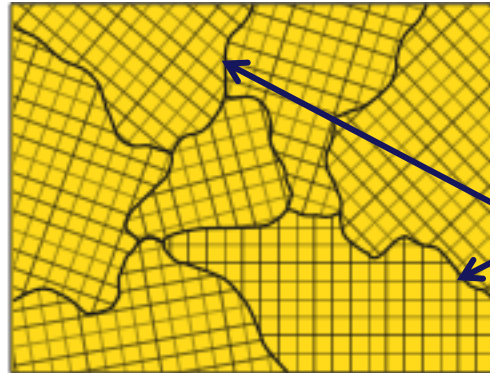
(a)



(b)



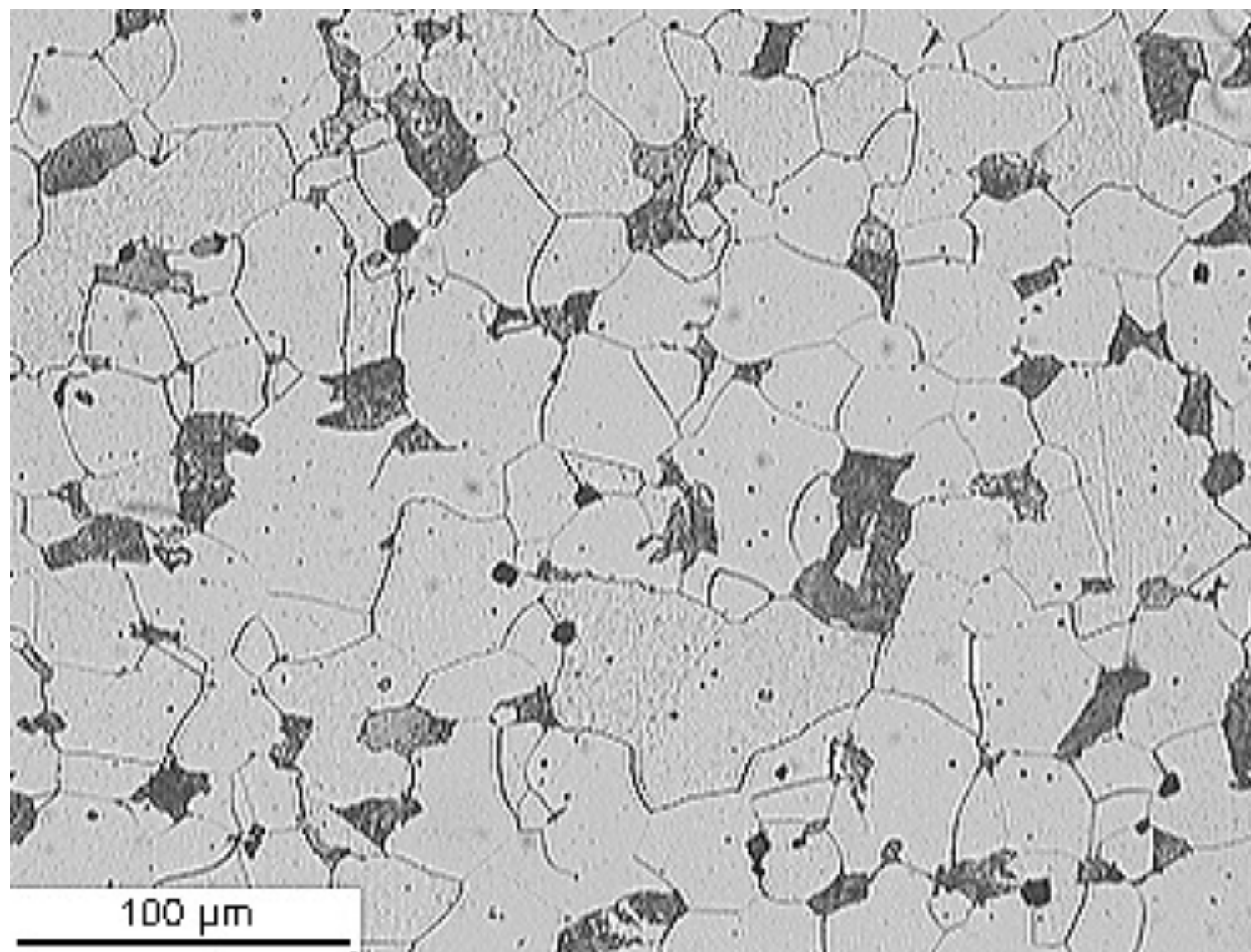
(c)



(d)

Grain
boundaries

Various stages during solidification of molten metal.

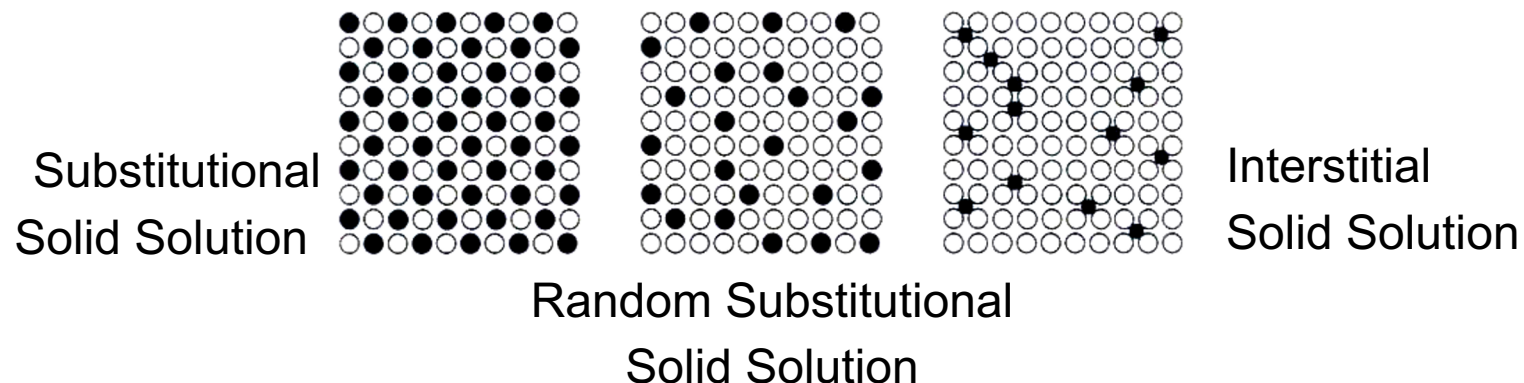




Alloy: composition of two or more chemical elements, at least one of which is metal.

Two basic forms:

- **Solid solutions:** A solute (minor element) is added to the solvent (major element). Substitutional and interstitial solid solutions are possible.
- **Intermetallic compounds:** Complex structures consisting of two metals in which solute atoms are present among solvent atoms in certain proportions.





Defined as a compound consisting of long-chain molecules, each molecule made up of repeating units connected together.

- The word *polymer* is derived from the Greek words *poly*, meaning many, and *meros* (reduced to *mer*), meaning part.
- There may be thousands, even millions of units in a single polymer molecule.
- Most polymers are based on carbon and are therefore considered organic chemicals.



Polymers can be separated into **plastics** and **rubbers**.

As engineering materials, it is appropriate to divide them into the following three categories:

1. Thermoplastic polymers;
2. Thermosetting polymers;
3. Elastomers;

where (1) and (2) are plastics and (3) are rubbers.



THERMOPLASTIC POLYMERS OR THERMOPLASTICS

Solid materials at room temperature, but viscous liquids when heated to temperatures of only a few hundred degrees:

- This characteristic allows them to be easily and economically shaped into products;
- They can be subjected to heating and cooling cycles repeatedly without significant degradation;
- Symbolized by TP.



THERMOSETTING POLYMERS OR THERMOSETS

Cannot tolerate repeated heating cycles as thermoplastics can:

- When initially heated, they soften and flow for molding (pre-polymers);
- Elevated temperatures also produce a chemical irreversible reaction that hardens the material into an infusible solid;
- If reheated, thermosets degrade, burn or char rather than soften;
- Symbolized by TS.



Polymers that exhibit extreme elastic extensibility when subjected to relatively low mechanical stress:

- Some elastomers can be stretched by a factor of 10 and yet completely recover to their original shape;
- Although their properties are quite different from thermosets, they share a similar molecular structure that is different from the thermoplastics.



Thermoplastics are commercially the most important of the three types:

- About 70% in weight of all synthetic polymers produced;
- Thermosets and elastomers share the remaining 30% about evenly, with a slight edge for the former.

On a volumetric basis, current annual usage of polymers exceeds that of metals.



REASONS WHY POLYMERS ARE IMPORTANT

Plastics can be molded into intricate part shapes, usually with no further processing:

- Very compatible with *net shape* processing.

On a volumetric basis, polymers:

- Are cost competitive with metals;
- Generally require less energy to produce than metals.

Certain plastics are transparent, which makes them competitive with glass in some applications.



Nearly all polymers used in engineering are synthetic:

- They are made by chemical processing.

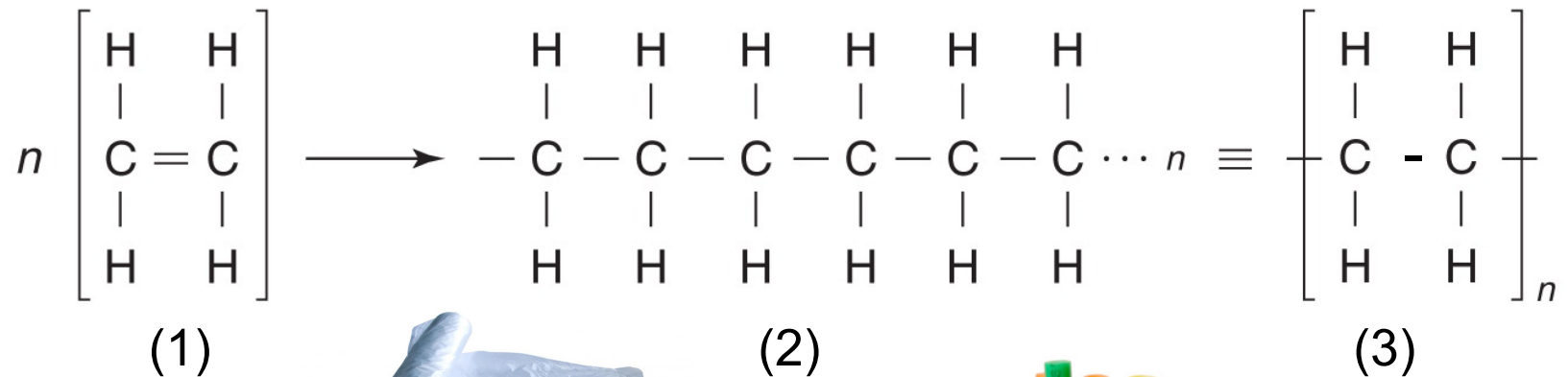
Polymers are synthesized by joining many small molecules together into very large molecules, called *macromolecules*, that possess a chain-like structure.

The small units, called *monomers*, are generally simple unsaturated organic molecules, such as ethylene C_2H_4 .



Synthesis of polyethylene from ethylene monomers:

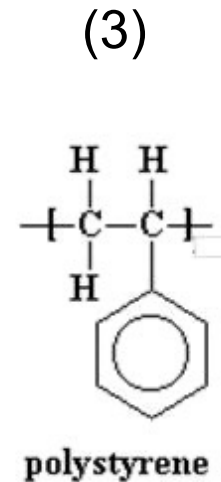
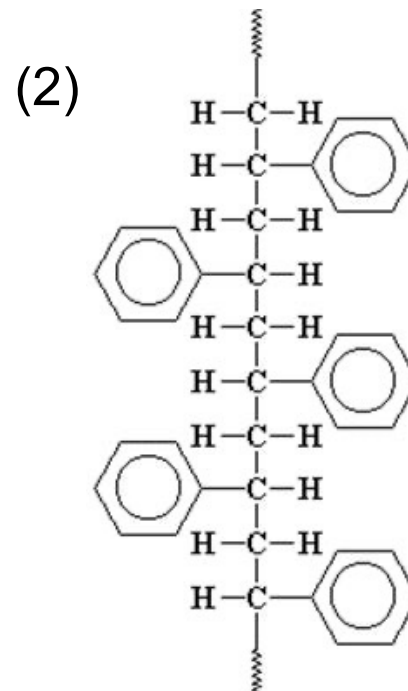
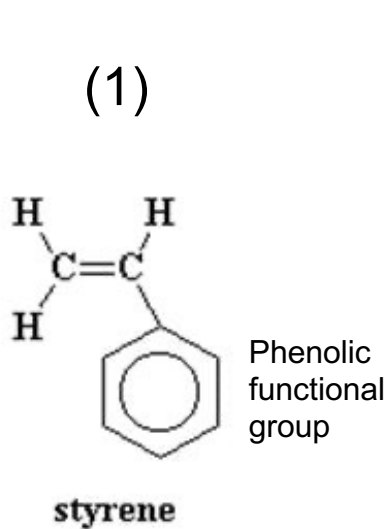
- 1) n ethylene monomers,
- 2) polyethylene of chain length n ;
- 3) concise notation for depicting polymer structure of chain length n .





Synthesis of polystyrene from styrene monomers:

- 1) n styrene monomers,
- 2) polystyrene of chain length n ;
- 3) concise notation for depicting polymer structure of chain length n .





Monomers can be linked into polymers in repeating units to make longer and larger molecules by a chemical process called a **polymerization reaction**.

Although there are several variations, two polymerization processes are important:

1. ***condensation*** polymerization;
2. ***addition*** polymerization.



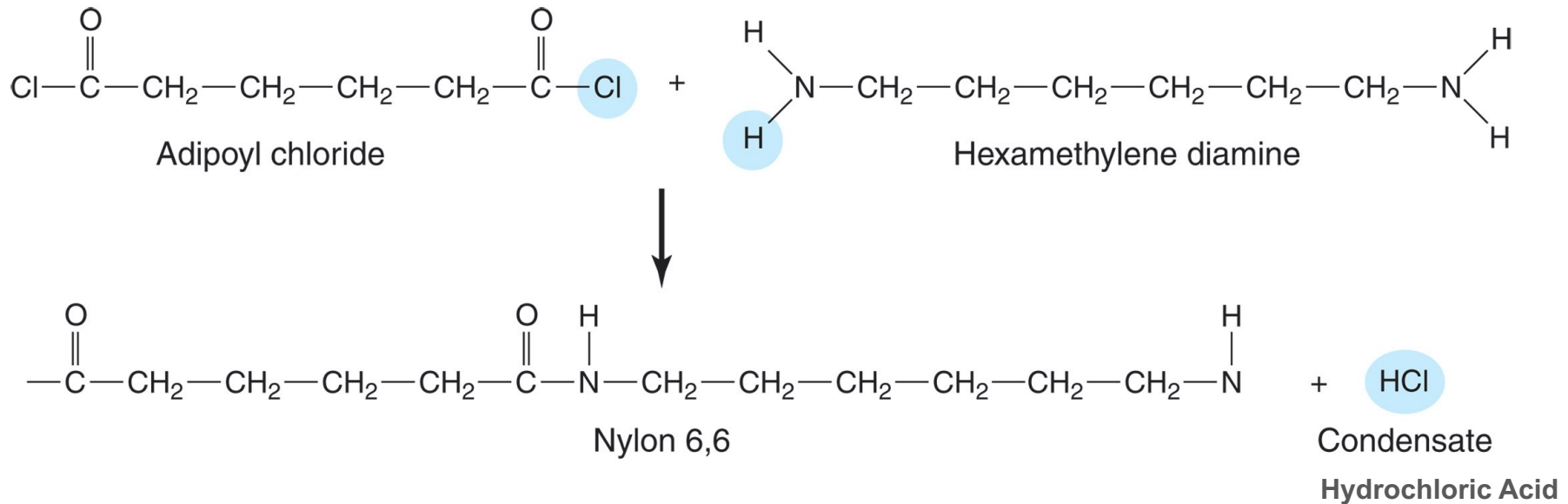
In ***condensation polymerization***, polymers are produced by the formation of bonds between two types of reacting mers.

A characteristic of this reaction is that *reaction by-products* (such as water H_2O , or hydrochloric acid HCl) are condensed out.

This process is also known as *step-growth* or *step-reaction polymerization*, because the polymer molecule grows step-by-step until all of one reactant is consumed.



Condensation polymerization of nylon 6,6





In ***addition polymerization*** (also called *chain-growth* or *chain-reaction polymerization*), bonding takes place without reaction by-products.

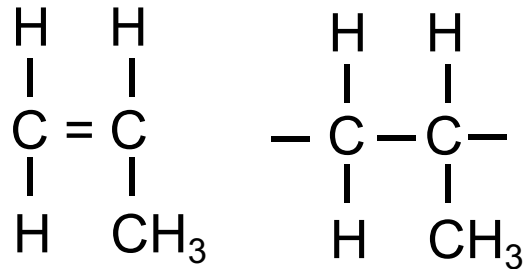
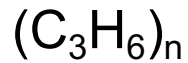
It is called “chain reaction” because of the high rate at which long molecules form simultaneously, usually within a few seconds.

This rate is much higher than that in condensation polymerization. In addition polymerization, an *initiator* is added to open the double bond between two carbon atoms, which begins the linking process by adding many more monomers to a growing chain.

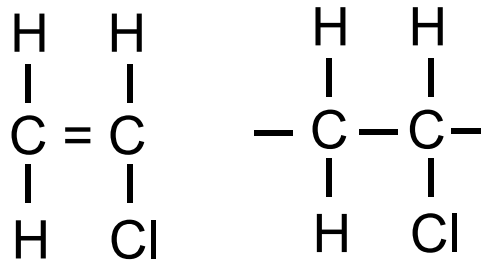
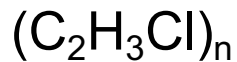


Addition polymerization

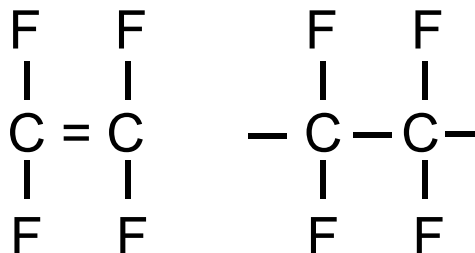
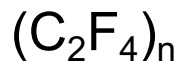
Polypropylene (PP)



Polyvinyl chloride (PVC)

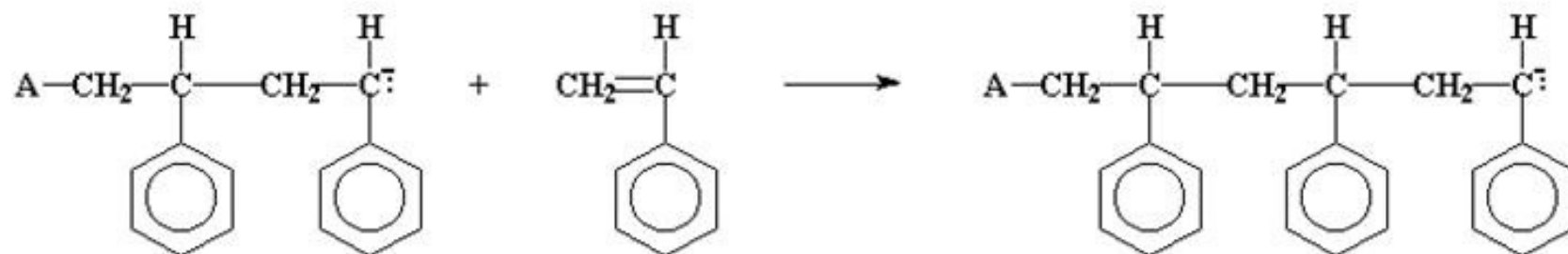
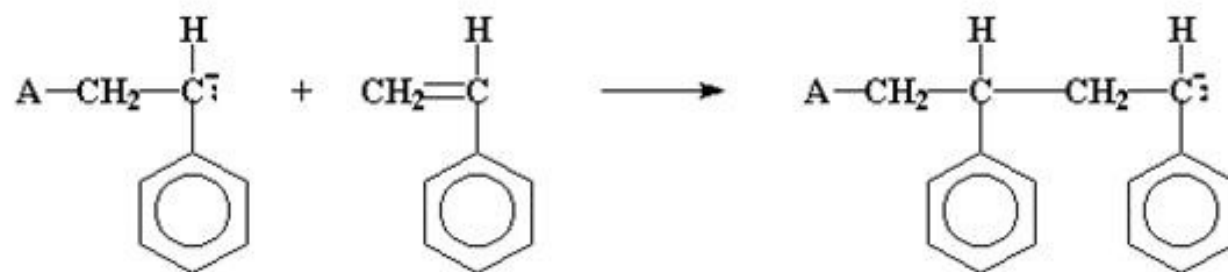
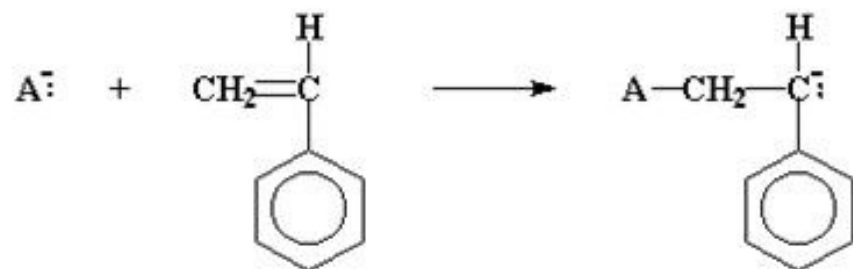


Polytetrafluoroethylene
(PTFE - Teflon)





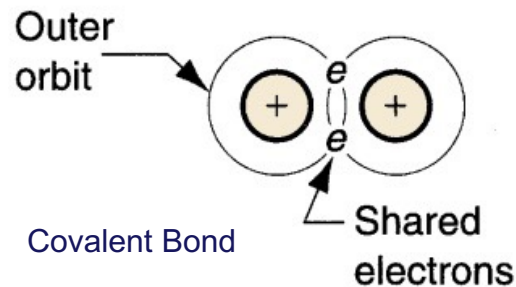
Anionic addition polymerization of styrene





STRUCTURE OF POLYMERS

During polymerization, the monomers are linked together by covalent bonds (a primary bond), forming a polymer chain. The polymer chains are, in turn, held together by secondary bonds, such as van der Waals bonds, hydrogen bonds, and ionic bonds.

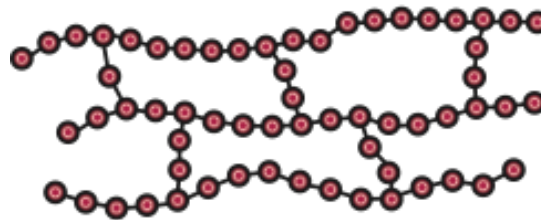
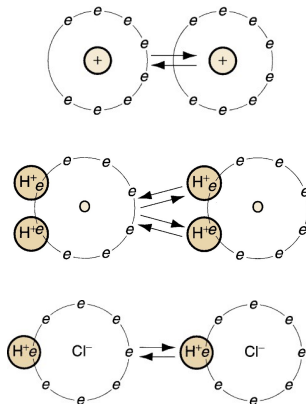


(a) Linear

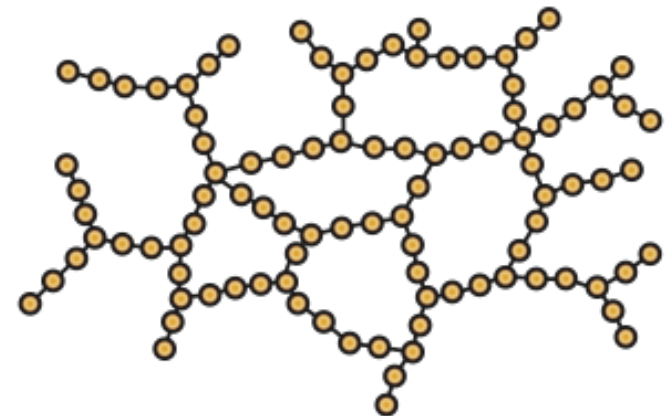


(b) Branched

Secondary Bonds



(c) Cross-linked



(d) Network

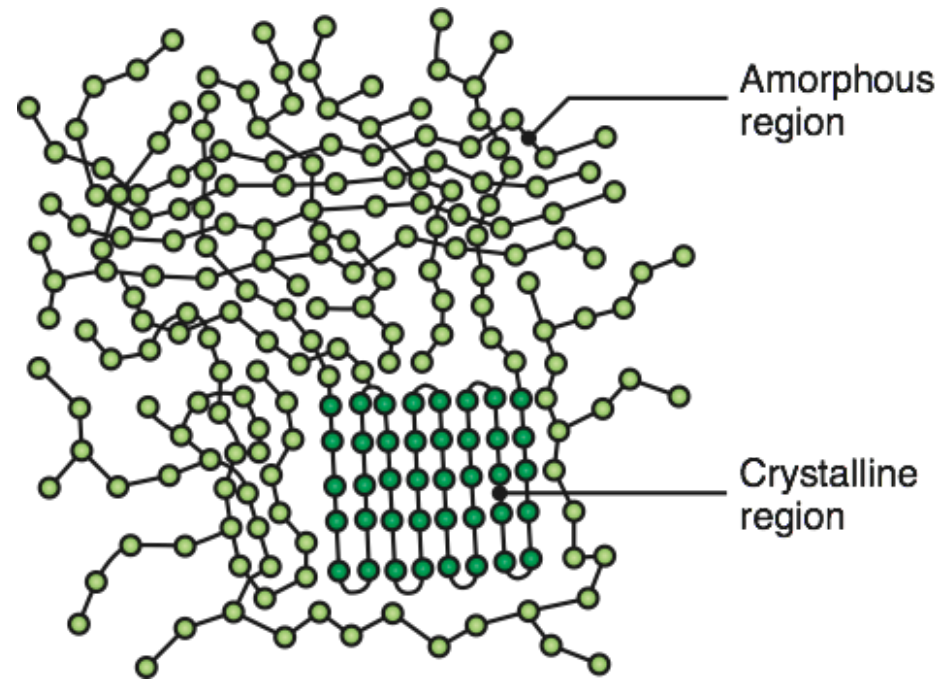
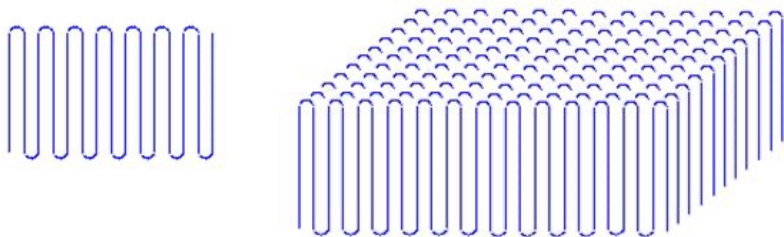


POLYMER CRYSTALLINITY

Polymers are generally **amorphous**. The amorphous arrangement of polymer chains often is described as being like a bowl of spaghetti, all intertwined with each other.

In some polymers, however, it is possible to impart some **crystallinity** and thereby modify their characteristics. This arrangement may be fostered either during the synthesis of the polymer or by deformation during its subsequent processing.

Degree of crystallinity: ratio between the weight of crystalline regions and the overall weight of the polymer (usually < 60 - 70%).





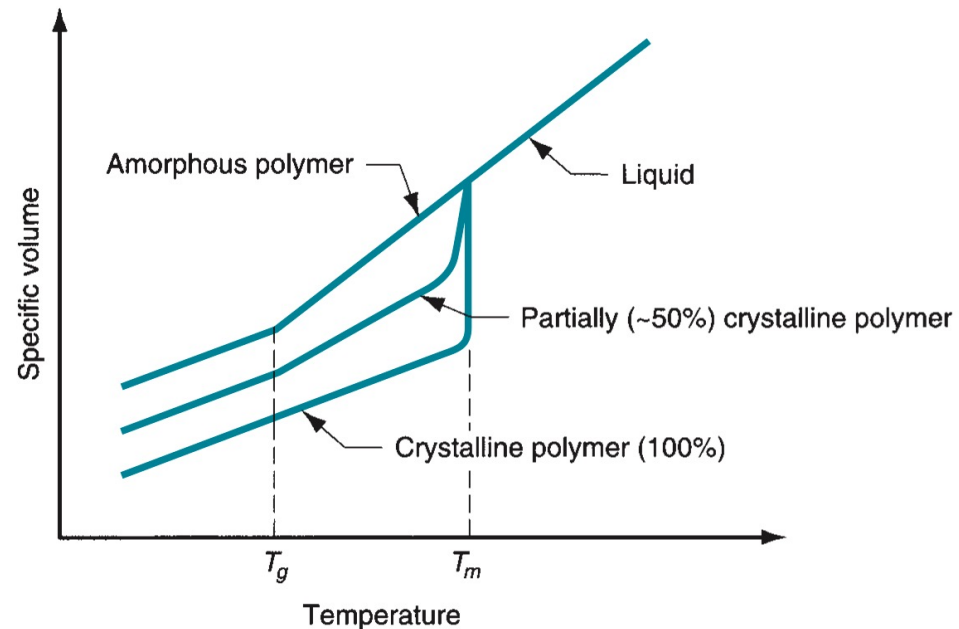
TRANSITION TEMPERATURES OF POLYMERS

100% crystalline polymer: sharp liquid-solid transition temperature (*melting temperature* - T_m).

Amorphous polymers do not have a specific melting point, they undergo a distinct change in their mechanical behaviour across a narrow range of temperatures.

At low temperatures, they are hard, rigid, brittle, and glassy; at high temperatures, they are rubbery or leathery.

The temperature at which a transition occurs is called the **glass-transition temperature** (T_g), also called the glass point or glass temperature.





TRANSITION TEMPERATURES OF POLYMERS

Glass-transition and Melting Temperatures of Some Polymers

Material	T_g (°C)	T_m (°C)
Nylon 6,6	57	265
Polycarbonate	150	265
Polyester	73	265
Polyethylene		
High density	−90	137
Low density	−110	115
Polymethylmethacrylate	105	—
Polypropylene	−14	176
Polystyrene	100	239
Polytetrafluoroethylene	−90	327
Polyvinyl chloride	87	212
Rubber	−73	—



THERMOPLASTIC POLYMERS (TP)

Thermoplastic polymers can be heated from solid state to viscous liquid and then cooled back down to solid:

- Heating and cooling can be repeated many times without degrading the polymer;
- Reason: TP polymers consist of linear and/or branched macromolecules that do not cross-link.

Thermosets and elastomers change chemically when heated, which cross-links their molecules and permanently cures these polymers.



THERMOSETTING POLYMERS (TS)

TS polymers are distinguished by their highly cross-linked three-dimensional, covalently-bonded structure.

Chemical reactions associated with cross-linking are called *curing* or *setting*.

In effect, formed part becomes a large macromolecule.

Always amorphous and exhibits no glass transition temperature.

Cannot be re-melted, instead they degrade, burn, or char.



CROSS-LINKING (CURING) OF TS POLYMERS

Three categories:

1. Temperature-activated systems: curing caused by heating;
2. Catalyst-activated systems: curing by small amounts of a catalyst added to the liquid polymer;
3. Mixing-activated systems: two starting ingredients react to cause curing.

Curing is done at the fabrication plants that make the parts rather than the supplier of the starting materials.



Polymers capable of large elastic deformation when subjected to relatively low stresses:

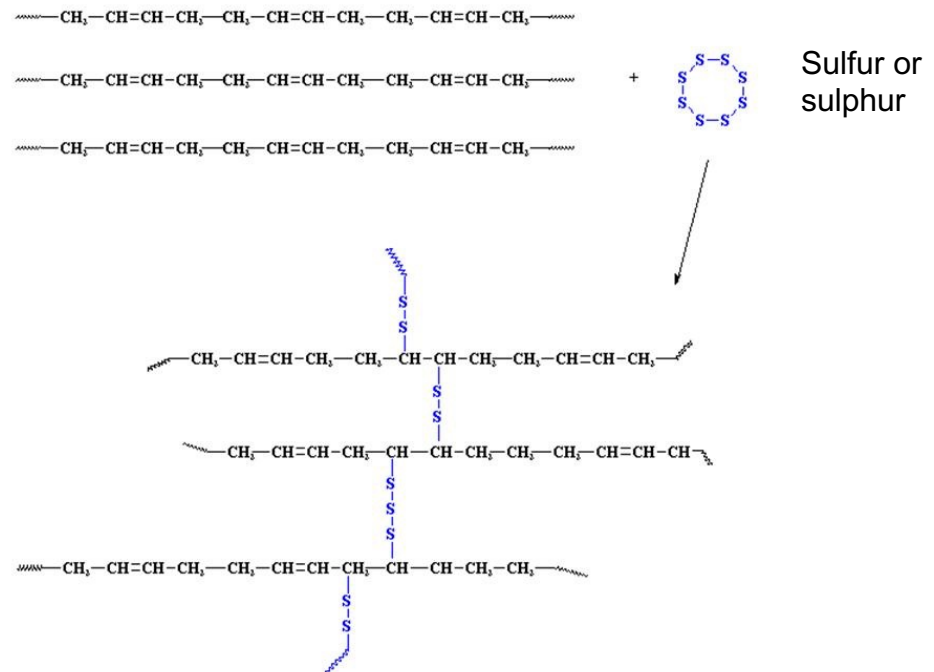
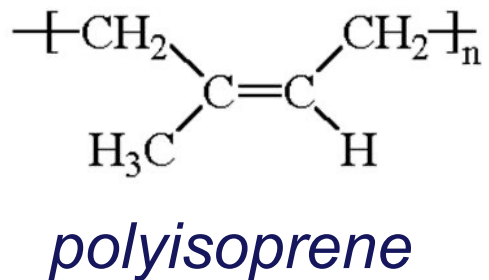
- Some can be extended 500% or more and still return to their original shape.
- Two categories:
 1. Natural rubber - derived from biological plants;
 2. Synthetic polymers - produced by polymerization processes like those used for TP and TS polymers.

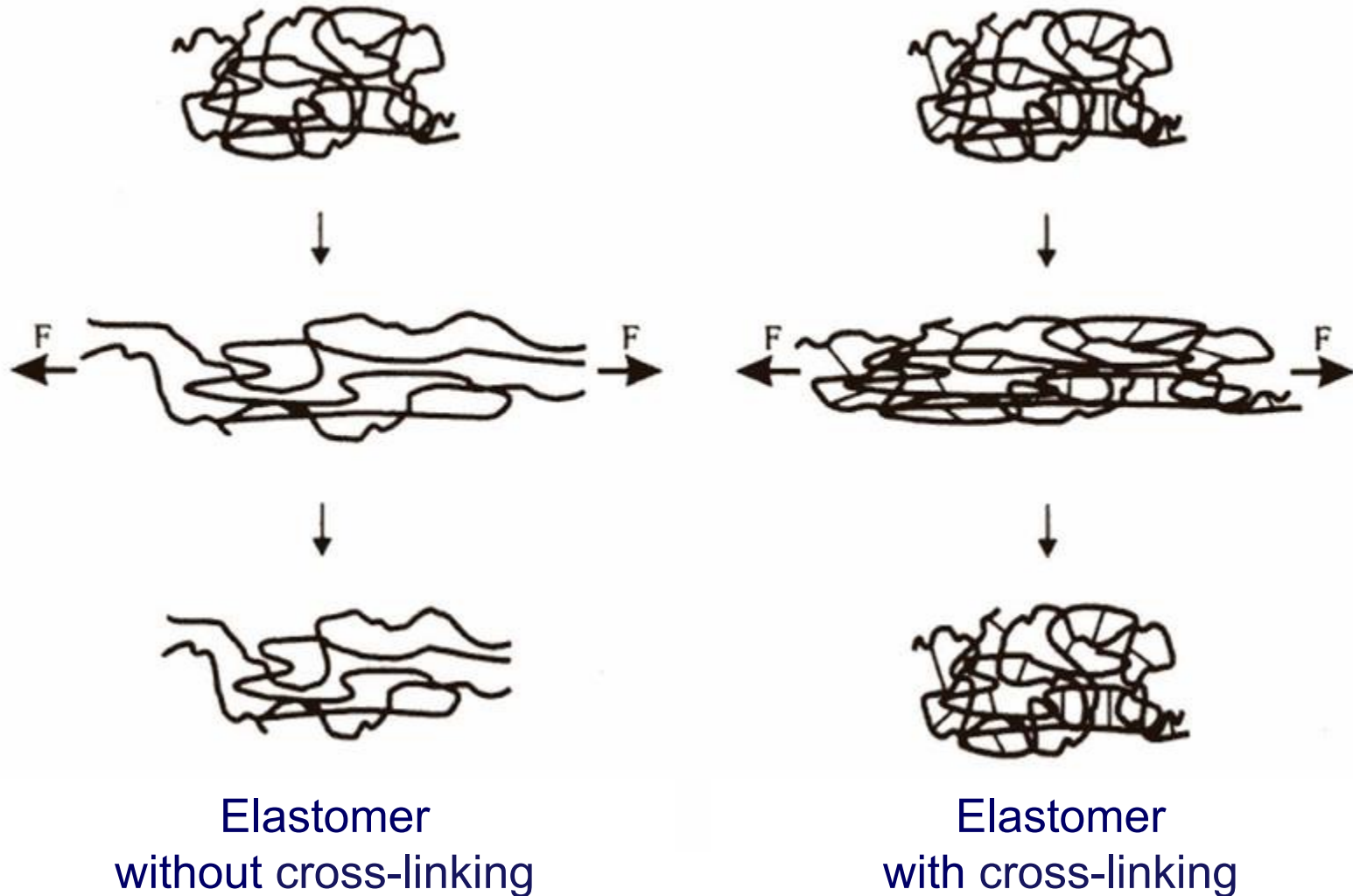


Vulcanization is the term for curing (cross-linking) in the context of natural rubber (and certain synthetic rubbers).

Typical cross-linking in rubber is one to ten links per hundred carbon atoms in the linear polymer chain, depending on degree of stiffness desired.

Considerable less than cross-linking in thermosets.







General Recommendations for Plastic Products

Design requirement	Typical applications	Plastics
Mechanical strength	Gears, cams, rolls, valves, fan blades, impellers, pistons	Acetals, nylon, phenolics, polycarbonates, polyesters, polypropylenes, epoxies, polyimides
Wear resistance	Gears, wear strips and liners, bearings, bushings, roller blades	Acetals, nylon, phenolics, polyimides, polyurethane, ultrahigh-molecular-weight polyethylene
Frictional properties		
High	Tires, nonskid surfaces, footwear, flooring	Elastomers, rubbers
Low	Sliding surfaces, artificial joints	Fluorocarbons, polyesters, polyimides
Electrical resistance	All types of electrical components and equipment, appliances, electrical fixtures	Polymethylmethacrylate, ABS, fluorocarbons, nylon, polycarbonate, polyester, polypropylenes, ureas, phenolics, silicones, rubbers
Chemical resistance	Containers for chemicals, laboratory equipment, components for chemical industry, food and beverage containers	Acetals, ABS, epoxies, polymethylmethacrylate, fluorocarbons, nylon, polycarbonate, polyester, polypropylene, ureas, silicones
Heat resistance	Appliances, cookware, electrical components	Fluorocarbons, polyimides, silicones, acetals, polysulfones, phenolics, epoxies
Functional and decorative	Handles, knobs, camera and battery cases, trim moldings, pipe fittings	ABS, acrylics, cellulose, phenolics, polyethylenes, polypropylenes, polystyrenes, polyvinyl chloride
Functional and transparent	Lenses, goggles, safety glazing, signs, food-processing equipment, laboratory hardware	Acrylics, polycarbonates, polystyrenes, polysulfones
Housings and hollow shapes	Power tools, housings, sport helmets, telephone cases	ABS, cellulose, phenolics, polycarbonates, polyethylenes, polypropylene, polystyrenes